Context Rot Is Two Mechanisms: Displacement Acts Through Attention Mass, Competition Does Not

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Abstract

Long conversations are said to wear language models down. The phenomenology is easy to elicit—as context fills, output entropy collapses several-fold and the last token's attention drifts off the system prompt and the current query onto recent text—but the causal claim behind it has never been tested inside a single harness, because the standard setups confound *accumulation* (more context) with *dilution* (the answer-bearing tokens becoming a smaller, more distant fraction of it). We separate them. Accumulating **individually localized** tasks, where the evidence for the current answer always sits at the current query, we reproduce the full signature battery across four model families and find **no accuracy cost** and no loss of instruction adherence. Holding items, model, fill and metric fixed and moving only *where the evidence lives*, the cost appears at once: displacing it costs 0.19–0.21 accuracy, and in a joint fit distance carries the coefficient ($\beta = -0.0076$ $[-0.0117, -0.0035]$) while fill carries none. We then show this cost is **causally mediated by attention mass**. Clamping the evidence's attention share at its original position, with nothing moved, reproduces **114%** of the displacement penalty ($+0.202$ $[+0.138, +0.267]$, landing indistinguishably on the displaced arm); a dose-response sweep over the share recovers the same effect to within 0.004 and shows the mass$\rightarrow$accuracy curve is smooth and graded, with no threshold anywhere in it. The drain itself is positional in *tokens*, not turns. Holding distance and fill fixed and varying only how *confusable* the accumulated context is isolates a *second and separate* cost: near-duplicate context is worth -0.085 $[-0.140, -0.030]$ while leaving the evidence's attention mass untouched (-0.0003 $[-0.0009, +0.0004]$, where the measured dose-response predicts 2% of the observed effect). So context rot is two mechanisms, not one—displacement acts through attention mass and competition does not—and needle benchmarks vary both at once, which is why they have not been told apart. Meanwhile the signatures are in-context comfort: the entropy collapse vanishes on organic WildChat dialogue and tracks conversation homogeneity, not length, and across OLMo-2's post-training chain it is a baseline-confidence level shift installed mostly by SFT. Context rot, as measured, is not a decay of competence with length: accumulation alone is free, and what costs accuracy is the evidence being displaced or contested—one of which is attention dilution and one of which is not.

1 Introduction

A recurring worry about long conversations is *context rot*: as turns accumulate, the model is said to degrade. The phenomenology is real. On a repeating multiple-choice task, instruction-tuned models become dramatically more confident as context fills—next-token entropy drops 3–4$\times$ on

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Table 1: **Distance is the variable that matters** (OLMo-2-Instruct, n=192 per arm, chance = 0.200, fill 0.688 in every arm). Evidence attention share is measured at layer 24 on the same forwards. Case-resampled bootstrap 95% CIs.

arm	`local`	`back_2`	`back_5`	`back_10`	`back_20`
accuracy	**0.464**	0.359	0.292	0.250	0.276
cost vs `local`	—	+0.104	+0.172	+0.214	+0.188
95% CI	—	$[+.005, +.203]$	$[+.073, +.266]$	$[+.120, +.307]$	$[+.094, +.281]$
evidence share @L24	0.0408	0.0363	0.0320	0.0195	0.0124

Qwen-2.5-7B and Llama-3.1-8B, in both question orders—and the last token's attention shifts systematically away from the system prompt and the current query toward the most recent prior cases, with the distribution diffusing (at layer 24 of OLMo-2-Instruct: system↔fill -0.93, recent↔fill $+0.89$, current↔fill -0.71, entropy↔fill $+0.95$, all monotone in fill). By every visual diagnostic this is a strong effect that grows with context.

The tempting reading is that the model is being worn down by length. But almost every context-rot evaluation embeds the answer-bearing information inside *one long task*—a long document, a buried needle, a growing codebase. There, longer context mechanically *dilutes* the relevant tokens among irrelevant ones, so a measured degradation conflates the cost of accumulation per se with the cost of the target being harder to attend to ("lost in the middle," Liu et al., 2023). These cannot be told apart in that setup, and the distinction matters practically: an agent transcript of finished, locally-scoped steps accumulates without diluting, and the two accounts make opposite predictions about it.

In this paper we separate them inside one harness. We accumulate **individually localized** tasks, so that context length and the dilution of the answer-bearing evidence become independent knobs, and we intervene on each in turn. Accumulation alone costs nothing. Moving the evidence k turns back, at identical fill and byte-identical questions, costs 0.19–0.21 accuracy (Table 1).

That cost is *causally mediated by attention mass*. A clamp that sets a span's post-softmax attention share lets us remove the evidence's mass while leaving it exactly where it is: doing so reproduces 114% of the displacement penalty, and a dose-response sweep over the share recovers the same endpoint to within 0.004 across two separately-run experiments.

A second cost is not mediated that way at all. Holding the evidence at the current query and varying only how *confusable* the accumulated context is, near-duplicate context costs a further 0.085 accuracy while the evidence's attention share does not move measurably—2% of what the measured dose-response would require. Taken together, context rot is two mechanisms with different signatures, and a needle-in-a-haystack benchmark varies both at once, which is why they have not been told apart.

Our contributions are as follows:

- A harness that makes accumulation and dilution independent, by accumulating individually localized tasks whose evidence never moves. In it, accumulation alone costs no accuracy and no instruction adherence, with bounded nulls rather than point estimates.
- A causal mediation result for displacement: removing the evidence's attention mass with nothing moved reproduces 114% of the displacement penalty, and the mass→accuracy relationship is smooth and graded with no threshold in it.
- A dissociation establishing a *second* mechanism: competition costs accuracy at constant attention mass, so "attention dilution" is the right account of displacement and the wrong account of competition.

2 Background

Length as the suspected cause. Most evidence for context rot comes from setups where the answer-bearing information sits inside a single long input. Liu et al. (2023) showed that accuracy depends strongly on *where* in the context the relevant passage sits, dipping when it falls in the middle. Levy et al. (2024) held a reasoning task fixed and padded the input with task-irrelevant text, finding degradation well before the models' advertised context limits. Hong et al. (2025) ran a broad sweep over 18 models and found that reliability falls with input length even on tasks as simple as retrieval and replication, and that the fall is not uniform: it depends on how similar the needle is to the question, and on whether distractors are present. Laban et al. (2025) report a large multi-turn degradation, but locate its cause in the model committing early to a wrong reading and failing to recover—not in length as such.

Confusable context as a separate suspect. A second literature manipulates the *content* of the surrounding context rather than its length. Shi et al. (2023) insert an irrelevant sentence into grade-school math problems and observe a sharp accuracy drop. In retrieval-augmented generation, Cuconasu et al. (2024) find that documents retrieved as relevant but not containing the answer act as distractors and degrade accuracy.

What is not separated. In each of these setups the two candidate causes move together. Padding a prompt lengthens it *and* pushes the evidence further from the query; adding a distractor document does the same while also supplying a competing answer. So a measured degradation cannot be assigned to accumulation, to the evidence becoming distant, or to the surrounding text becoming confusable. A mechanistic thread—Liu et al. (2023)'s positional account, and attention-based explanations generally—suggests attention as the channel, but attention is measured alongside these manipulations rather than intervened on. This paper separates the three factors inside one harness and intervenes on attention directly.

3 Preliminaries

Localized task streams. Our harness accumulates **individually localized tasks**: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the *current* question, which always sits at the end of the context; prior turns are never required to answer it. Accumulation therefore grows the context without ever diluting the information the current answer needs. A reversed-question-order control removes difficulty-ordering confounds. Because a 4–32k window holds only tens of cases, we pool many independent accumulation sessions.

The dilution knob. The design's real value is that dilution becomes an *independent knob*. Holding the item, the model, the total fill and the metric fixed, we can move the evidence k user turns back into the accumulated transcript, with the question text byte-identical across arms and the same filler in every arm. Distance and fill, welded together in a long-document setup, move independently here. All causal experiments run on OLMo-2-7B-Instruct at layer 24, seed 42, with an overflow guard that skips rather than truncates every item that would not fit whole—truncating a long item near the window edge manufactures exactly the late-window cost under study.

The attention-mass clamp. To intervene on attention rather than position, we add a clamp that *sets* a token span's post-softmax attention share: a constant bias on the span's key columns scales its odds, and softmax renormalizes, so nothing in the attention forward is reimplemented and a scale of 1.0 is bit-identical to the unhooked model. The clamp is uniform across heads.

Analysis. Per-condition n is bounded by the context window, so every clamp experiment is paired over shared items and analysed with a paired bootstrap resampling *cases* (10,000 draws), never heads or turns.

4 Results

4.1 Accumulation

Per-case accuracy does not fall as context fills; it is worst at the cold start and warms up ($\mathrm{corr}(\text{correct}, \text{fill}) = +0.11$ Instruct, $+0.07$ DPO). The obvious objection is that a homogeneous repeating task lets in-context learning mask a hidden decay, so we remove that crutch: in a *random-subject* stream each question is drawn uniformly from all 56 MMLU subjects, so prior context carries no predictive information about the next question—only the answer *format* is learnable. Accuracy is still flat (random: $\mathrm{corr} = -0.02\ [-0.10, +0.05]$, n=699; coherent single-subject: $\mathrm{corr} = +0.01$, n=1001). These nulls are *bounded*: the 95% interval on upper-minus-lower-half accuracy excludes

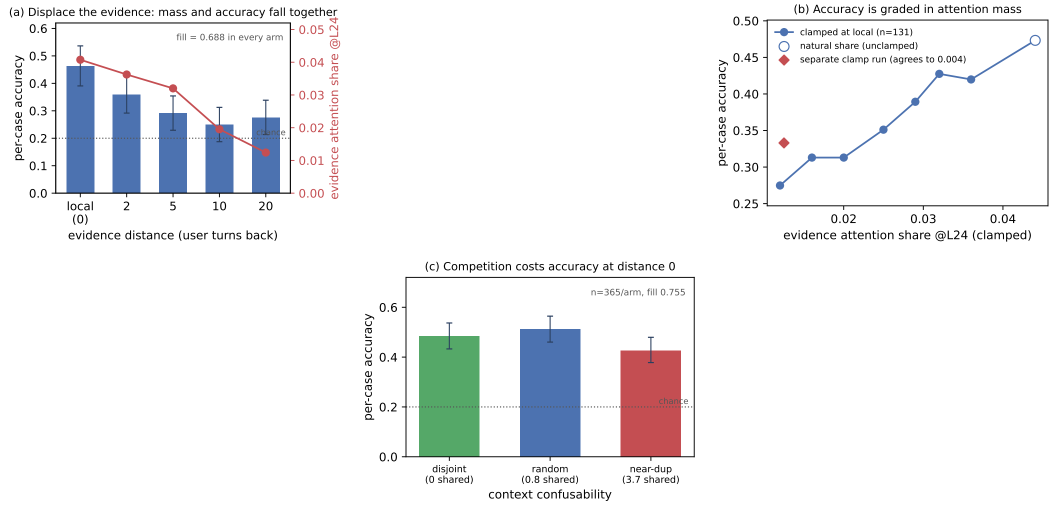

Figure 1: **Dilution costs accuracy; accumulation does not.** (a) Moving the evidence k user turns back, at identical fill and byte-identical questions, drains its attention mass and costs accuracy together. (b) Setting that mass directly, with the evidence left in place, reproduces the penalty and traces a smooth graded dose-response—no threshold. The diamond is an independently-run clamp experiment measuring the same endpoint cut; the two agree to 0.004. (c) At distance 0 and fixed fill, near-duplicate accumulated context costs a further 0.085. Bootstrap CIs resample cases throughout.

declines beyond 4.1 points (coherent) and 9.4 (random). Checkable instruction adherence—a canary system instruction to begin every reply with a marker and end with a tag—stays at ceiling throughout ($\mathrm{corr}(\text{violation}, \text{fill}) = 0$), an honest null that bounds rather than proves, since the canaries were easy.

4.2 Displacement

We now vary only *where the answer-bearing evidence lives*. Each DDXPlus probe's vignette is placed k user turns before its question; mean fill is 0.688 in every arm, the question text is byte-identical, and the overflow guard skipped 0 of 192 probes. Every gap's CI excludes zero (Table 1, Fig. 1a). In a joint fit of accuracy on fill and distance, **distance carries the coefficient** ($\beta = -0.00761$ $[-0.01173, -0.00346]$, significant) and **fill carries none** ($\beta = -0.00725$ $[-0.21006, +0.18658]$). Within the `local` arm—the one that reproduces the paper's original design—accuracy is flat with fill ($\beta = -0.294$ $[-0.767, +0.184]$). The effect survives restriction to parsed responses and is then fully monotone ($0.524 \to 0.413 \to 0.397 \to 0.343 \to 0.338$). One honest qualification: the decline saturates by $k{\approx}10$ rather than deepening to $k{=}20$, and those two arms are not distinguishable from each other. So the same model, on the same items at the same fill, degrades when the evidence is displaced and does not degrade when it is not. The localized null is not a ceiling artifact: these items *can* show degradation, and do, as soon as dilution is reintroduced.

Distance and attention drain move together by construction, so Table 1 is consistent with dilution but does not establish it—a purely positional account predicts the same table. Four interventions separate them.

Observational correlations. Evidence share falls $0.0408 \to 0.0124$ across the ladder ($r = -0.83$ with distance) while the *question*'s share stays flat, confirming only the evidence moved. A trap worth recording: *within* an arm, higher evidence share predicts *lower* accuracy ($\beta = -11.2$ $[-20.0, -2.6]$, stronger controlling for vignette length). Attention there tracks item difficulty—the model looks harder at confusing vignettes—and has the *opposite sign* to the causal effect below. Anyone reading share-against-accuracy correlations off this data without intervening will conclude the reverse of the truth.

Mass removal. Keeping the evidence at `local` and clamping its share down to the *same item's own* `back_20` share costs $+0.202$ $[+0.138, +0.267]$ (paired $n{=}174$) and lands at 0.333 against `back_20`'s 0.365—a difference of -0.025 $[-0.083, +0.035]$, indistinguishable. **Mass removal alone reproduces 114% of the displacement penalty, with nothing moved.**

Dose-response. Sweeping the evidence's share at fixed position over seven levels (common subset n=131 present at every level) gives a monotone decline from 0.473 at the natural share (0.0441) to 0.275 at 0.012 (Fig. 1b). Six of the seven levels differ significantly from the natural share—including a cut of less than a fifth of the mass—and no step is a threshold: the largest adjacent step is 0.053 and the decline is smooth throughout. The endpoint contrast, $+0.198$ $[+0.115, +0.282]$, matches the independently-run clamp experiment's $+0.202$ $[+0.138, +0.267]$ for the same share change—agreement to 0.004 across two separate runs, which is the strongest internal consistency check in the program. An earlier draft of this work read a *threshold* into two flat contrasts; the sweep refutes it. The relationship is *graded*: a measured slope, not a cliff.

Mass restoration. The converse is only partial: clamping the displaced evidence's share back *up* to its `local` level recovers a real but small 32% of the penalty ($+0.055$ $[+0.017, +0.098]$), leaving a significant residual ($+0.123$ $[+0.054, +0.193]$). Our uniform clamp can strip mass faithfully but cannot reconstruct *which* heads should carry the restored mass. Full sufficiency with partial necessity is the signature of an instrument that is faithful in one direction and crude in the other, and we report it as one rather than as a second mechanism. A pattern-matched, per-head clamp is the test that would settle it.

Tokens versus turns. A partial 2×2 in (gap turns) $\times$ (filler length), total context equalized by padding, dissociates them: two arms matched on *tokens* have effectively identical evidence share (0.0104 vs 0.0108) despite a $4\times$ difference in turn count, while at matched turns share falls $0.0288 \rightarrow 0.0104$ as the gap grows $446 \rightarrow 1684$ tokens—a 64% cut in the evidence's attention that costs at most 9.4 accuracy points ($+0.026$ $[-0.042, +0.094]$). Share regresses significantly on gap tokens and not on gap turns. Attention drain is a function of token distance, as RoPE decay predicts; conversation structure is not what drains it.

The current query. Accumulation drives the *current query*'s share from ≈ 0.35 down to ≈ 0.15, and it is natural to ask whether that stops short of a floor. It does not. Clamping the query span on cold-start contexts, accuracy is flat from the natural share (0.258, accuracy 0.545) down through 0.20 (0.509), then costs 16.4 points at 0.15 ($+0.164$ $[+0.036, +0.291]$)—which is exactly the share accumulation reaches. Accumulation does not stop far short of the shoulder; it stops *at* it. (Levels below 0.15 require -4.7 to -6.1 nats, near-ablation rather than points on a dose-response—the model answers "A" $110/110$—and are excluded from interpretation.) That accuracy nonetheless stays flat under accumulation, while the same share reached by clamping costs points, is itself evidence that in-context learning is actively holding accuracy up.

4.3 Competition

Burying a needle varies two things at once: the evidence becomes distant *and* it becomes surrounded by plausible alternatives. So far we have isolated distance. We isolate the other half by holding the evidence at the current query and fill fixed, and varying only how *confusable* the accumulated context is. Every arm accumulates eight prior DDXPlus cases, each answered in the transcript with its gold letter, so format and in-context-learning affordance are constant; the arms differ only in how much the accumulated cases' differentials overlap the current probe's (`disjoint` 0 shared options, `random` 0.80, `near_dup` 3.75 of 5, always with a *different* gold, so the context never displays the probe's answer as a correct one). Paired over n=365 probes, accuracy is `random` 0.512, `disjoint` 0.485, `near_dup` $\mathbf{0.427}$.

Near-duplicate context costs -0.085 $[-0.140, -0.030]$ against the natural stream and -0.058 $[-0.112, -0.006]$ against disjoint context; in a joint fit within this experiment, **option overlap carries the coefficient** ($\beta = -0.021$ $[-0.039, -0.003]$) and **fill carries none** ($\beta = -0.284$ $[-0.640, +0.075]$)—the same shape as the distance result, on a different variable. Two controls matter: the low-competition arms must land on §4.2's `local` accuracy of 0.464, and both do ($+0.049$ $[-0.037, +0.134]$ and $+0.021$ $[-0.065, +0.107]$); and context length does not order the arms, since `disjoint` is the *longest* and mid-accuracy while `near_dup` is the shortest, so length biases against the result. Restricted to parsed responses only, `random` $-$ `near_dup` $= +0.082$ $[+0.025, +0.138]$ survives. The effect is *not* monotone in confusability: `random` sits above `disjoint`, not below it, though not significantly ($+0.027$ $[-0.019, +0.074]$). Mild topical overlap plausibly supplies useful priming over the option space while near-duplication supplies competitors that actively mislead; only

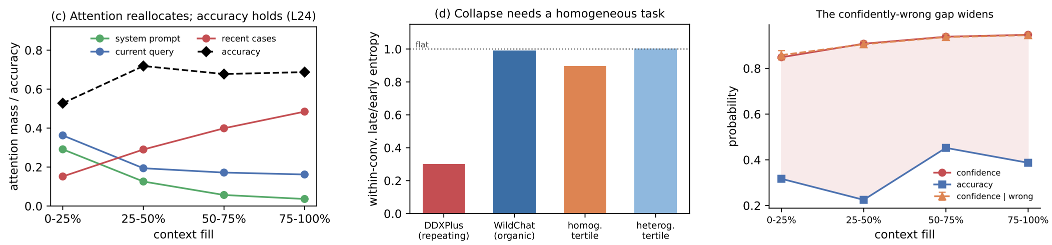

Figure 2: **The signatures are in-context comfort, and the residual hazard is calibration.** (a) Attention reallocates monotonically off the system prompt and current query with fill while accuracy holds. (b) The entropy collapse appears only with a homogeneous repeating task, not on organic WildChat dialogue, and tracks homogeneity rather than length. (c) Confidence rises with fill on wrong answers as much as right, at flat accuracy.

the second is large enough to measure here. We therefore claim that near-duplicate context is costly, *not* that cost rises with overlap—one significant step, not a gradient.

Attention mass under competition. Repeating the sweep with attention capture instead of generation (same probes, same seeds) shows the headline contrast moves accuracy 8.5 points with **no detectable change in the evidence's attention mass**: `random` $-$ `near_dup` $= -0.00027$ $[-0.00088, +0.00035]$ in evidence share at layer 24. The dose-response above is 6.29 accuracy per unit share (R^2=0.97), so that share difference predicts an accuracy gap of 0.0017—**2% of what is measured**, and 6.5% even at the CI's upper end. Reproducing this cost through mass would need a share change 50× larger. The two contrasts even run opposite ways: `random` $-$ `disjoint` has a significant *share* difference and no accuracy difference, while `random` $-$ `near_dup` has a significant *accuracy* difference and no share difference. **Displacement costs accuracy by draining attention mass; competition costs accuracy while leaving it intact.** "Attention dilution" is the right account of the first and the wrong account of the second—and a needle-in-a-haystack benchmark varies both at once, which is precisely why they have not been told apart before.

4.4 Confidence signatures

If the entropy collapse were a generic depth effect it should survive on heterogeneous dialogue. It does not. On 150 organic WildChat conversations (Zhao et al., 2024) (Qwen-2.5-7B, teacher-forced over the real history), within-conversation entropy is flat with depth: median late/early ratio 0.99, median within-conversation $\mathrm{corr}(\text{entropy}, \text{depth})$ -0.02, and 52% of conversations trend down—a coin flip. Partitioning WildChat by its *own* inter-turn homogeneity isolates the driver: the entropy slope correlates with homogeneity (partial $r = -0.151$ controlling for length), homogeneous conversations collapse (0.897 late/early) while heterogeneous ones are flat (1.001), and because homogeneous conversations are *longer*, length works against the effect.

Replaying one probe across OLMo-2-7B's (OLMo Team, 2024) `base`→`SFT`→`DPO`→ `Instruct` chain (identical cases and orders) shows early-context entropy collapsing monotonically, 1.06 → 0.58 → 0.33 → 0.20, with the largest single drop at base→SFT. But the "collapses *as context fills*" framing is mostly a base-model ICL effect plus a floor: the within-context ratio *falls* across stages (1.64 → 0.47), because aligned models start near a confidence floor. Post-training is a downward *level shift* in baseline entropy, not a steeper collapse slope. Finally, the mechanistic story proposed for the collapse does not survive a causal test: clamping the boundary-detector SAE feature (Gemma Scope F90871, Lieberum et al., 2024) back to its clean level on held-out fatigued cases moves entropy the *wrong* way (0.442 → 0.465 vs. clean 0.293), leaves accuracy flat, and a random feature of similar magnitude perturbs entropy at least as much.

Consistently, conditioning on fill, the cases OLMo-2 answers *wrong* have *higher* current-query attention than the ones it gets right (pooled $\Delta = +0.045$ $[-0.003, +0.097]$, n=115; sign positive in every fill bin and at every probed layer). Errors are associated with *fixating* on the query; correct answers spread attention onto the accumulated cases—the per-case fingerprint of the in-context learning that keeps accuracy up. The "rot" and the within-context predictor of a correct answer point in opposite directions.

5 Conclusion and Outlook

Limitations. Per-condition n is bounded by the context window, so every interval here depends on the paired analysis of §3; resampling the arms independently inflates each one about $2.5\times$, enough to report a real effect as null. The necessity direction of the mediation is only partially established and is plausibly instrument-limited, since the clamp is uniform across heads. The competition result is one significant step rather than a gradient—mild overlap is indistinguishable from none, and we do not claim cost rises with confusability. The instruction-canary null is a floor effect, the WildChat figures are summary values from teacher-forced analyses, and the causal program runs on one model family. The accuracy null holds for localized tasks; genuinely length-dependent single-task settings are where dilution is the point, and are out of scope by design.

With accumulation, displacement and competition separated inside one harness: accumulation produces a real, monotone reallocation of attention and a real collapse of *confidence*, but no loss of accuracy and no loss of instruction adherence. Displacing the evidence produces a large accuracy cost that is causally mediated by its attention mass, graded in that mass, and positional in tokens. Making the context confusable produces a further cost that is *not* mediated by that mass at all. The literature's central claim is therefore *correct but misattributed, and under-counted*: context rot is not a decay with length, it is displacement and competition, two mechanisms with different signatures, of which length is merely the usual way of causing both at once.

The residual hazard in the localized case is *calibration*, not competence: on the DDXPlus streams logging per-token confidence (n=154 pooled cases, two models, both question orders), answer confidence rises $0.85 \to 0.95$ with fill ($r = +0.72$ $[+0.64, +0.79]$) at flat accuracy—and equally on *wrong* answers ($0.86 \to 0.95$; $r = +0.69$ $[+0.57, +0.78]$)—so the confidently-wrong gap widens by $\sim$10 points over a session (Fig. 2c), most consequential exactly where a homogeneous high-stakes stream makes the model surest. The test we would run next is the pattern-matched, per-head clamp that would settle whether the partial necessity above is an instrument limitation or a second channel within displacement itself.

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